

Computer Graphics

Project #1

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Throughout the project we used [Three.js](#) and WebGL, a couple algorithms that we used to create this project were THREE.Shape and THREE.ShapeGeometry. These allowed for us to make the shapes and allow for the shapes to be customized given size and color. We also used raycasting to detect when the mouse clicked on a shape and was dragging it to a certain location. This also detected if the shape was close enough to its correct location for it to snap into place. A challenge that we faced was getting raycasting to work properly and allowing for the user to drag each shape, also allowing for pieces to snap into place took a while to figure out as at first pieces would snap in from almost any distance making the puzzle too easy, creating a function to only snap into place from a certain distance took some time to figure out.

One of the two extra functions we added was a gameplay timer in the top right corner of the screen. When the page is loaded in, the timer starts so the players can keep track of their time. When the puzzle is completed, the timer stops, allowing the player to see how long it took them to complete the puzzle.

The second extra feature included a victory animation screen when the puzzle is complete. When all the puzzle pieces have been correctly placed, a pop up screen shows up saying "Puzzle Complete! Time: ()". The puzzle constantly spins in a clockwise position as another visual feature. This creates a sense of accomplishment when completing the puzzle and adds a better visual experience for the user.